NETWORK PROBLEM 1 :

IP ADDRESS CONFLICT (LAYER 3 NETWORK LAYER)

IP Address Conflict Issue occurs when two or more devices on a network try to use the same IP address. This situation can lead to communication disruptions between devices on the network, connection drops, and even complete unavailability of some devices.

Causes of IP Address Conflict:

1. **Static IP Address Assignment:**
   - When a device in a network uses a static IP address, and another device manually assigns the same IP address, a conflict arises. o For example, if a computer in the network is assigned a static IP of 192.168.1.10, and another device tries to use the same IP, a conflict will occur.
2. **Dynamic IP Address Assignment (DHCP):**
   - The **DHCP server** assigns dynamic IP addresses to devices. If the DHCP server gives the same IP to a device that was already manually assigned a static IP or reuses IP addresses incorrectly from its pool, a conflict can occur. o In most networks, DHCP automatically assigns IP addresses, but misconfigurations in the process can lead to IP conflicts.
3. **DHCP Server Issues:**
   - If there are multiple DHCP servers on a network and they do not share the IP pool, conflicts can arise. For example, if one DHCP server assigns IPs from 192.168.1.100 to 192.168.1.200 while another DHCP server uses the same range, conflicts will occur.
4. **New Devices Added to the Network:**
   - When a device's IP address is dynamically assigned, a previously used but now removed device's IP address could be reassigned to a new device, causing an IP conflict.
5. **Hardware Failures in the Network:**
   - Hardware failures, particularly with the DHCP server, can also lead to IP conflicts. Similarly, misconfigurations in network cards or routers can cause conflicts.

Symptoms of IP Address Conflict:

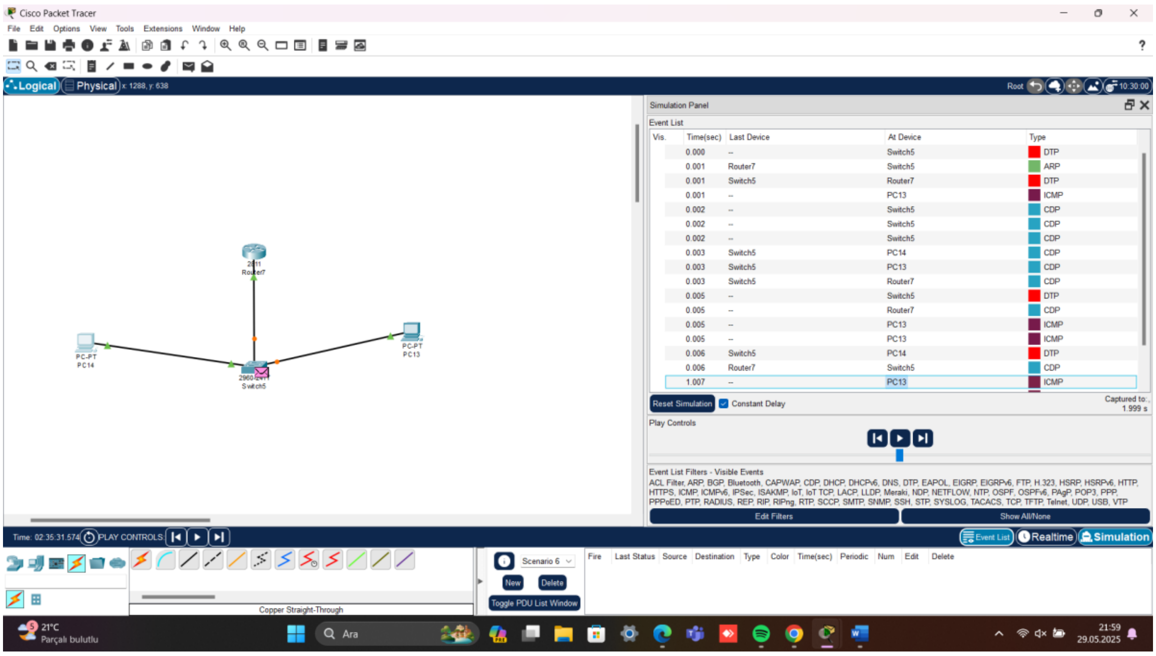
1. **Connection Drops:**
   - When a conflict occurs, devices on the network may fail to communicate properly, leading to dropped connections or very slow network speeds.
2. **Error Messages from Network Devices:**
   - When a conflict is detected, some devices may display error messages. For instance, Windows devices might show error messages regarding network adapter IP conflicts.
3. **Slow Performance:**
   - Networks with IP conflicts may experience slower performance as devices cannot send packets to each other properly, increasing network congestion.
4. **Network Services Not Working:**
   - IP conflicts can cause network services to fail. For example, file sharing services or printers may become inaccessible due to the conflict.

Resolving IP Address Conflict:

1. **Review Static IP Addresses:**
   - When assigning static IP addresses to devices in the network, make sure they do not overlap with the DHCP pool. For example, assign static IPs in the range from 192.168.1.1 to 192.168.1.10, and let the other devices automatically receive IP addresses via DHCP.
   - Proper **IP Address Planning** helps prevent conflicts on the network.
2. **DHCP Server Configuration:**
   - The DHCP server's IP pool should be correctly configured. The IP range assigned by the DHCP server should not overlap with the static IPs used in the network. Also, ensure that only one device manages the DHCP server.
3. **Detecting IP Conflicts in the Network:**
   - Devices with IP conflicts should be identified by network administrators, and their IP addresses should be manually changed. This can be done by checking the **ARP cache** (Address Resolution Protocol table) or using **ping** commands across the network.

4. **DHCP Lease Duration:**
   - The **DHCP lease duration** should be properly set. The lease duration determines how long a device can hold onto its assigned IP address. Shorter lease times can help release dynamic IPs more quickly and reduce conflict risks.
5. **Monitor Network Routers and Browsers:**
   - By reviewing router log files, error messages related to conflicts can be detected. The router interface will show IP assignments and potential conflict details.
6. **Update Network Hardware:**
   - Regular software updates for network hardware, especially routers and switches, are important. These updates ensure that the devices operate with the most secure and stable configurations.
7. **Switch from Static IP to Dynamic IP:**
   - If IP conflicts occur frequently, switching from static IP usage to dynamic IPs (DHCP) can help. This ensures that devices automatically receive IP addresses from the DHCP server, reducing the chances of conflicts.

Simulation of problem: In this simulation, pc13 and pc14 were given the same ip address. In this way, an ip conflict was simulated. The orange circles on the ports also indicate this.



Simulation of solution : Pc13 and Pc14 have different ip adresses (statically) on this simulation. In this scenario ,there is not ip conflict. We pinged pc14's ip address from pc13. It seems to work correctly in the command prompt.